JSS MAHAVIDYAPEETHA

**JSS SCIENCE AND TECHNOLOGY UNIVERSITY**

**Scheme of Teaching and Examination 2024-25**

**Outcome Based Education (OBE) and Choice Based Credit System (CBCS)**

(Effective from the academic year 2024-25)

SCHEME OF TEACHING AND ASSESSMENT FOR B.E

**Semester I: Physics cycle**

| Sl. No. | Course code | Course Title | Category code | Teaching Department | QP Setting Dept. | Teaching Hours per Week | | | | Credits | Marks | | | Exam Duration in hours |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | L | T | P | Total | | CIE | SEE | Total | |
| 1 | 24MA110 | Engineering Mathematics – I | BSC | MA | MA | 3 | 1 | - | 4 | 4 | 40 | 60 | 100 | 03 |
| 2 | 24PH110 | Engineering Physics | BSC | PH | PH | 3 | - | 2 | 5 | 4 | 40 | 60 | 100 | 03 |
| 3 | 24EE110 | Elements of Electrical Engineering | ESC | EE | EE | 3 | - | - | 3 | 3 | 40 | 60 | 100 | 03 |
| 4 | 24ME11x | Elective 1 – Mechanical Engineering | ESC | ME | ME | 3 | - | - | 3 | 3 | 40 | 60 | 100 | 03 |
| 5 | 24CS11x | Elective 2 - Introduction to Programming | ESC | CS/IS | CS/IS | 3 | - | 2 | 5 | 4 | 40 | 60 | 100 | 03 |
| 6 | 24HU110 | Kannada | HSMC | HU (KA) | HU (KA) | 2 | - | - | 2 | 0 | 40 | - | 40 | - |
| 7 | 24HU120 | Innovation & Design Thinking | HSMC | PD | HU | 2 | - | - | 2 | 1 | 20 | 30 | 50 | 02 |
| 8 | | NSS/NCC/SDA | | | | | | | | - | - | - | - | - |
| **Total** | | | | | | | | | | **19** | **260** | **330** | **590** | - |

| **Elective 1 – Mechanical Engineering** | |
|---|---|
| 24ME111 | Elements of Mechanical Engineering |
| 24ME112 | Mechatronics and Robotics |

| **Elective 2 - Introduction to Programming** | |
|---|---|
| 24CS111 | C Programming |
| 24CS112 | Python Programming |
| 24CS113 | Object Oriented Programming using JAVA |

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SCHEME OF TEACHING AND ASSESSMENT FOR B.E

**Semester I: Chemistry cycle**

| SlNo | Course code | Course Title | Category code | Teaching Dept. | QP Setting Dept. | Teaching Hours per Week | | | | Credits | Marks | | | Exam Duration in hours |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | L | T | P | Total | | CIE | SEE | Total | |
| 1 | 24MA110 | Engineering Mathematics –I | BSC | MA | MA | 3 | 1 | - | 4 | 4 | 40 | 60 | 100 | 03 |
| 2 | 24CH110 | Engineering Chemistry | BSC | CH | CH | 3 | - | 2 | 5 | 4 | 40 | 60 | 100 | 03 |
| 3 | 24CV110 | Engineering Mechanics | ESC | CV | CV | 3 | - | - | 3 | 3 | 40 | 60 | 100 | 03 |
| 4 | 24EC11x/ 24EI11x | Elective 3 – Electronics Engineering | ESC | EC/EI | EC/EI | 3 | - | - | 3 | 3 | 40 | 60 | 100 | 03 |
| 5 | 24ME120 | Engineering Graphics and Design | ESC | ME | ME | - | 1 | 2 | 3 | 2 | 40 | 60 | 100 | 03 |
| 6 | 24HU11x | Elective 4 – Sustainability/Skill | HSMC | All Dept | HU | 2 | - | - | 2 | 2 | 20 | 30 | 50 | 02 |
| 7 | 24HU130 | Communication English | HSMC | HU (EN) | HU (EN) | 2 | - | - | 2 | 1 | 20 | 30 | 50 | 02 |
| 8 | | NSS/NCC/SDA/Sports/Yoga | | | | | | | | - | - | - | - | - |
| **Total** | | | | | | | | | | **19** | **240** | **360** | **600** | - |

| **Elective 3 – Electronics Engineering** | |
|---|---|
| 24EC111 | Elements of Electronics Engineering |
| 24EC112 | Fundamentals of Robotics and its Applications |
| 24EC113 | Sensors and Measurements |
| 24EC114 | Internet of Everything |
| 24EI115 | Transducers and Measuring Systems |

| **Elective 4 – Sustainability/Skill** | |
|---|---|
| 24HU111 | Sustainability Engineering |
| 24HU112 | Introduction to Cyber Security |
| 24HU113 | Human Psychology |
| 24HU114 | Web Technology |
| 24HU115 | Biology for Engineers |
| 24HU116 | Carbon Sequestration |

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**Outcome Based Education (OBE) and Choice Based Credit System (CBCS)**

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SCHEME OF TEACHING AND ASSESSMENT FOR B.E

**Semester II: Physics cycle**

| SlNo | Course code | Course Title | Category code | Teaching Department | QP Setting Dept. | Teaching Hours per Week | | | | Credits | Marks | | | Exam Duration in hours |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | L | T | P | Total | | CIE | SEE | Total | |
| 1 | 24MA210 | Engineering Mathematics – II | BSC | MA | MA | 3 | 1 | - | 4 | 4 | 40 | 60 | 100 | 03 |
| 2 | 24CH210 | Engineering Chemistry | BSC | CH | CH | 3 | - | 2 | 5 | 4 | 40 | 60 | 100 | 03 |
| 3 | 24CV210 | Engineering Mechanics | ESC | CV | CV | 3 | - | - | 3 | 3 | 40 | 60 | 100 | 03 |
| 4 | 24EC21x<br>24EI21x | Elective 3 – Electronics Engineering | ESC | EC/EI | EC/EI | 3 | - | - | 3 | 3 | 40 | 60 | 100 | 03 |
| 5 | 24ME220 | Engineering Graphics and Design | ESC | ME | ME | - | 1 | 2 | 3 | 2 | 40 | 60 | 100 | 03 |
| 6 | 24HU21x | Elective 4 – Sustainability/Skill | HSMC | All Dept | HU | 2 | - | - | 2 | 2 | 20 | 30 | 50 | 02 |
| 7 | 24HU230 | Communication English | HSMC | HU (EN) | HU (EN) | 2 | - | - | 2 | 1 | 20 | 30 | 50 | 02 |
| 8 | | NSS/NCC/SDA/Sports/Yoga | | | | | | | | - | - | - | - | - |
| **Total** | | | | | | | | | | **19** | **240** | **360** | **600** | |

| **Elective 3 – Electronics Engineering** | |
|---|---|
| 24EC111 | Elements of Electronics Engineering |
| 24EC112 | Fundamentals of Robotics and its Applications |
| 24EC113 | Sensors and Measurements |
| 24EC114 | Internet of Everything |
| 24EI115 | Transducers and Measuring Systems |

| **Elective 4 – Sustainability/Skill** | |
|---|---|
| 24HU211 | Sustainability Engineering |
| 24HU212 | Introduction to Cyber Security |
| 24HU213 | Human Psychology |
| 24HU214 | Web Technology |
| 24HU215 | Biology for Engineers |
| 24HU216 | Carbon Sequestration |

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**Outcome Based Education (OBE) and Choice Based Credit System (CBCS)**

(Effective from the academic year 2024-25)

SCHEME OF TEACHING AND ASSESSMENT FOR B.E

**Semester II: Chemistry cycle**

| Sl. No. | Course code | Course Title | Category code | Teaching Department | QP Setting Dept. | Teaching Hours per Week | | | | Credits | Marks | | | Exam Duration in hours |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | L | T | P | Total | | CIE | SEE | Total | |
| 1 | 24MA210 | Engineering Mathematics – II | BSC | MA | MA | 3 | 1 | - | 4 | 4 | 40 | 60 | 100 | 03 |
| 2 | 24PH210 | Engineering Physics | BSC | PH | PH | 3 | - | 2 | 5 | 4 | 40 | 60 | 100 | 03 |
| 3 | 24ME21x | Elective 1 – Mechanical Engineering | ESC | ME | ME | 3 | - | - | 3 | 3 | 40 | 60 | 100 | 03 |
| 4 | 24EE210 | Elements of Electrical Engineering | ESC | EE | EE | 3 | - | - | 3 | 3 | 40 | 60 | 100 | 03 |
| 5 | 24CS21x | Elective 2 – Introduction to Programming | ESC | CS/IS | CS/IS | 3 | - | 2 | 5 | 4 | 40 | 60 | 100 | 03 |
| 6 | 24HU210 | Kannada | HSMC | HU (KA) | HU (KA) | 2 | - | - | 2 | 0 | 40 | - | 40 | - |
| 7 | 24HU220 | Innovation & Design Thinking | ESC | PD | HU | 2 | - | - | 2 | 1 | 20 | 30 | 50 | 02 |
| 8 | | NSS/NCC/SDA/Sports/Yoga | | | | | | | | - | - | - | - | - |
| **Total** | | | | | | | | | | **19** | **260** | **330** | **590** | - |

| **Elective 1 – Mechanical Engineering** | |
|---|---|
| 24ME211 | Elements of Mechanical Engineering |
| 24ME212 | Mechatronics and Robotics |

| **Elective 2 - Introduction to Programming** | |
|---|---|
| 24CS211 | C Programming |
| 24CS212 | Python Programming |
| 24CS213 | Object Oriented Programming using JAVA |

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**Outcome Based Education (OBE) and Choice Based Credit System (CBCS)**

(Effective from the academic year 2024-25)

SCHEME OF TEACHING AND ASSESSMENT FOR B.E

**Semester III:**

| Sl. No. | Course Code | Course Title | Category Code | Teaching Department | QP Setting Dept. | Teaching Hours per Week | | | | Credits | Marks | | | Exam Duration in hours |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | L | T | P | Total | | CIE | SEE | Total | |
| 1 | 24MATDIP310 | Foundations of Engineering Mathematics-I | BSC | MA | MA | 3 | 0 | 0 | 3 | - | - | 60 | 60 | 03 |
| 2 | 24MA313 | Engineering Mathematics - III | BSC | MA | MA | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 3 | 24CS310 | Object Oriented Programming using C++ | PCC | CS | CS | 3 | 0 | 2 | 5 | 4 | 40 | 60 | 100 | 03 |
| 4 | 24CS320 | Digital System Design | PCC | CS | CS | 3 | 0 | 2 | 5 | 4 | 40 | 60 | 100 | 03 |
| 5 | 24CS330 | Data Structures | PCC | CS | CS | 4 | 0 | 0 | 4 | 4 | 40 | 60 | 100 | 03 |
| 6 | 24CS340 | Computer Organization | PCC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 7 | 24CS350 | Discrete Mathematical Structures | PCC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 8 | 24CS36L | Data Structures Lab | PCCL | CS | CS | 0 | 0 | 3 | 3 | 1.5 | 40 | 60 | 100 | 03 |
| 9 | 24HU311 | Universal Human Values-II | HSMC | CS | HU | 2 | 0 | 0 | 2 | 2 | 20 | 30 | 50 | 02 |
| | | NSS/NCC/SD/Sports/ Yoga | SDA | | | | | | | - | - | - | - | - |
| **Total** | | | | | | | | | | **24.5** | **300** | **450/ 510*** | **750/ 810*** | |

*For lateral entry student's total marks is 810

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SCHEME OF TEACHING AND ASSESSMENT FOR B.E

**Semester IV:**

| Sl. No. | Course Code | Course Title | Category Code | Teaching Department | QP Setting Dept. | Teaching Hours per Week | | | | Credits | Marks | | | Exam Duration in hours |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | L | T | P | Total | | CIE | SEE | Total | |
| 1 | 24MATDIP410 | Foundations of Engineering Mathematics-II | BSC | MA | MA | 3 | 0 | 0 | 3 | 0 | - | 60 | 60 | 03 |
| 2 | 24MA413 | Linear Algebra | BSC | MA | MA | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 3 | 24CS410 | Operating Systems | PCC | CS | CS | 3 | 0 | 2 | 5 | 4 | 40 | 60 | 100 | 03 |
| 4 | 24CS420 | Design and Analysis of Algorithms | PCC | CS | CS | 4 | 0 | 0 | 4 | 4 | 40 | 60 | 100 | 03 |
| 5 | 24CS430 | Data Communication | PCC | CS | CS | 4 | 0 | 0 | 4 | 4 | 40 | 60 | 100 | 03 |
| 6 | 24CS440 | Software Engineering | PCC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 7 | 24CS45xOE | Open Elective-I | OEC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 8 | 24CS46L | Design and Analysis of Algorithms Lab | PCCL | CS | CS | 0 | 0 | 3 | 3 | 1.5 | 40 | 60 | 100 | 03 |
| 9 | 24HU412 | Environmental Studies | HSMC | EV | EV | 2 | 0 | 0 | 2 | 2 | 20 | 30 | 50 | 02 |
| | | NSS/NCC/SD/Sports/Yoga | SDA | | | | | | | - | - | - | - | - |
| **Total** | | | | | | | | | | **24.5** | **300** | **450/ 510*** | **750/ 810*** | |

*For lateral entry student's total marks is 810

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SCHEME OF TEACHING AND ASSESSMENT FOR B.E

**Semester V:**

| Sl. No. | Course Code | Course Title | Category Code | Teaching Department | QP Setting Dept. | Teaching Hours per Week | | | | Credits | Marks | | | Exam Duration in hours |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | L | T | P | TOTAL | | CIE | SEE | Total | |
| 1 | 24CS510 | Computer Networks | PCC | CS | CS | 3 | 0 | 2 | 5 | 4 | 40 | 60 | 100 | 03 |
| 2 | 24CS520 | Database Management Systems | PCC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 3 | 24CS530 | Theory of Computation | PCC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 4 | 24CS540 | Artificial Intelligence | PCC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 5 | 24CS55x | Professional Elective-I | PEC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 6 | 24CS56x | Professional Elective-II | PEC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 7 | 24CS57xOE | Open Elective-II | OEC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 8 | 24CS58L | Database Management Systems Lab | PCCL | CS | CS | 0 | 0 | 3 | 3 | 1.5 | 40 | 60 | 100 | 03 |
| | | NSS/NCC/SD/Sports/Yoga | SDA | | | | | | | - | - | - | - | - |
| **Total** | | | | | | | | | | **23.5** | **320** | **480** | **840** | |

*Details provided in the syllabus

| Professional Elective-I | | Professional Elective-II | |
|---|---|---|---|
| 24CS551 | Image Processing and Computer Vision | 24CS561 | JAVA and J2EE |
| 24CS552 | Data Science and Applications | 24CS562 | Computer Architecture |
| 24CS553 | Multimedia Computing | 24CS563 | Software Testing and Quality Assurance |

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SCHEME OF TEACHING AND ASSESSMENT FOR B.E

**Semester VI:**

| Sl. No. | Course code | Course title | Category Code | Teaching Department | QP Setting Dept. | Teaching Hours per Week | | | | Credits | Marks | | | SEE Duration in hours |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | L | T | P | TOTAL | | CIE | SEE | Total | |
| 1 | 24CS610 | Internet of Things | PCC | CS | CS | 3 | 0 | 2 | 5 | 4 | 40 | 60 | 100 | 03 |
| 2 | 24CS620 | Machine Learning | PCC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 3 | 24CS630 | Principles of Compilers | PCC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 4 | 24CS640 | Cryptography and Network Security | PCC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 5 | 24CS65x | Professional Elective-III | PEC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 6 | 24CS66x | Professional Elective-IV | PEC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 7 | 24CS67xOE | Open Elective-III | OEC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 8 | 24CS68L | Machine Learning Lab | PCCL | CS | CS | 0 | 0 | 3 | 3 | 1.5 | 40 | 60 | 100 | 03 |
| 9 | 24HU610 | Constitution of India and Professional Ethics | HSMC | HU | - | 2 | 0 | 0 | 2 | 0 | 40 | - | 40 | - |
| | | NSS/NCC/SD/Sports/Yoga | SDA | | | | | | | - | - | - | - | - |
| **Total** | | | | | | | | | | **23.5** | **360** | **480** | **840** | |

*Details provided in the syllabus

| **Professional Elective-III** | | **Professional Elective-IV** | |
|---|---|---|---|
| 24CS651 | Cloud Computing | 24CS661 | Blockchain Technology |
| 24CS652 | Automotive Cyber Security | 24CS662 | Natural Language Processing |
| 24CS653 | Cyber Security | 24CS663 | Full Stack Development |
| 24CS654 | Data Mining and Warehousing | 24CS664 | Robotics |

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SCHEME OF TEACHING AND ASSESSMENT FOR B.E

**Semester VII:**

| Sl. No. | Course code | Course title | Category Code | Teaching Department | QP Setting Dept. | Teaching Hours per Week | | | | Credits | Marks | | | SEE Duration in hours |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | L | T/ SDA | P | TOTAL | | CIE | SEE | Total | |
| 1 | 24CS710 | Software Project Management | HSMC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 2 | 24CS72x | Professional Elective-V | PEC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 3 | 24CS73x | Professional Elective-VI | PEC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 4 | 24CS74P | Industrial training /Internship | PWC | CS | - | - | - | - | - | 2 | 40 | - | 40 | - |
| 5 | 24CS75P | Project Work Phase – I | PWC | - | - | - | - | - | - | 2 | 40 | - | 40 | - |
| **Total** | | | | | | | | | | **13** | **200** | **180** | **380** | |

*Details provided in the syllabus

| **Professional Elective-V** | | **Professional Elective-VI** | |
|---|---|---|---|
| 24CS721 | Deep Learning | 24CS731 | Object Oriented Modeling and Design |
| 24CS722 | DevOps | 24CS732 | Text and Web Mining |
| 24CS723 | Big Data Analytics | 24CS733 | Enterprise Resource Planning |

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SCHEME OF TEACHING AND ASSESSMENT FOR B.E

**Semester VIII:**

| Sl. No. | Subject Code | Course title | Category Code | Teaching Department | QP Setting Dept. | Teaching Hours per Week | | | | Credits | Marks | | | SEE Duration in hours |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | L | T | P | TOTAL | | CIE | SEE | Total | |
| 1 | 24CS81x | Professional Elective-VII | PEC | CS | CS | 3 | 0 | 0 | 3 | 3 | 40 | 60 | 100 | 03 |
| 2 | 24CS82P | Project work Phase – II | PWC | CS | CS | - | - | - | - | 10 | 40 | 60 | 100 | 03 |
| **Total** | | | | | | | | | | **13** | **80** | **120** | **200** | |

*Details provided in the syllabus

| **Professional Elective-VII** | |
|---|---|
| 24CS811 | Game Theory |
| 24CS812 | High Performance Computing |
| 24CS813 | Quantum Computing |

# III Semester

| DEPARTMENT | Mathematics | | | | | | |
|---|---|---|---|---|---|---|---|
| *Course Code* | 24MATDIP310 | *Total Credits* | 00 | *Course Type* | Theory | | |
| *Course Title* | **Foundations of Engineering Mathematics-I** (For III Semester Lateral Entry Students) | | | | | | |
| *Teaching Learning Process* | | *Contact Hours* | *Credits* | *Assessment in Weightage and marks* | | | |
| | *Lecture* | 03 | 00 | | CIE | SEE | Total |
| | *Tutorial* | 00 | 00 | *Weightage* | -- | 100% | 100% |
| | *Practical* | -- | -- | *Maximum Marks* | -- | 60 | 60 Marks |
| | *Total* | 03 | 00 | *Minimum Marks* | -- | 24 | 24 Marks |

**Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)**

**Course Prerequisite:** Basics in differential calculus, Integral calculus and Vectors.

**Course Objectives:**

| Sl. No. | Course Objectives |
|---|---|
| 1 | Know the behavior of the polar curve and its application. |
| 2 | Learn the notation of partial differentiation and determine the value of the Jacobians. |
| 3 | Learn how a function of single variable can be expanded as a Taylor's series and Maclaurin series and handle the indeterminate form. |
| 4 | Learn how the vectors govern the physical models. |
| 5 | Know how the real word problems governed by the first order differential equations. |

**Course Outcomes (COs)**

| CO | Course Outcomes | Highest Level of Cognitive Domain |
|---|---|---|
| CO1 | Apply the knowledge of calculus to solve problems related to polar curves and its applications in determining the amount of bending of the curve. | L2 |
| CO2 | Learn the notation of partial differentiation to calculate rates of change of multivariate functions and solve problems related to composite functions. | L2 |
| CO3 | Understand how a function of single variable can be expanded as a Taylor's series and handle the indeterminate form. | L2 |
| CO4 | Illustrate the applications of multivariate calculus to understand the characteristics of vector field. | L2 |
| CO5 | Apply the analytical methods to solve first order and first-degree differential equations and solve some Engineering problems. | L2 |

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

**Course Content / Syllabus:**

| Unit No. | Content | Hours Lecture |
|---|---|---|
| 1 | **Differential Calculus-I:** Polar curves: angle between the radius vector and tangent, angle of intersection of polar curves. Pedal equation for polar curves. Curvature and radius of curvature in pedal form. (Without proof) | 08 |
| 2 | **Differential Calculus–II:** Partial differentiation: Partial derivatives, Total derivatives, differentiation of composite functions. Jacobians. | 08 |
| 3 | **Differential Calculus–III:** Taylor's and Maclaurin's series expansions for one variable **(statement only).** Indeterminate forms, L' Hospital Rule (without proof) (Only $\frac{0}{0}, \frac{\infty}{\infty}, 0 \times \infty \; and \; \infty - \infty \; form$) | 08 |
| 4 | **Vector Calculus:** Vector Differentiation: scalar and vector fields, Gradient, directional derivative, divergence, curl-physical interpretation; solenoidal and irrotational vector fields-illustrative problems. | 07 |
| 5 | **Differential Equations-I:** Solution of first order and first degree differential equations: Variable Separable, Linear differential equations and Exact differential equations. | 08 |

**Text Books:**

| Sl. No. | Author/s | Title | Publisher Details |
|---|---|---|---|
| 1 | B. S. Grewal | Higher Engineering Mathematics | 44th Edition, Khanna Publications, 2015. |
| 2 | Erwin Kreyzig | Advanced Engineering Mathematics | 10th edition, Wiley Publications, 2016. |
| 3 | M.K. Jain, S.R.K. Iyenger and R.K. Jain | Numerical methods for scientific and engineering computation | 6th Edition, New Age International Publishers, 2012. |

**Reference Books:**

| Sl. No. | Author/s | Title | Publisher Details |
|---|---|---|---|
| 1 | N. P. Bali and Manish Goyal | A Textbook of Engineering Mathematics | Vol. I & II (AICTE), 10th Edition, Laxmi Publications Pvt Ltd, 2019. ISBN: 9789352743766, 9789352743766. |
| 2 | Maurice D. Weir, Joel R. Hass and George B. Thomas | Thomas' calculus: Early Transcendentals | 12th Edition, Pearson Education, 2016. ISBN: 978-07802-426-9 |

| | | | |
|---|---|---|---|
| 3 | B. V. Ramana | Higher Engineering Mathematics | 11th Edition, Tata-McGraw Hill 2010. ISBN: 0-07-053516-7. |
| 4 | Peter V.O 'Neil | Advanced Engineering Mathematics | 7th Edition, CENGAGE Learning India Pvt. Ltd. Publishers, 2012. ISBN: 978-81-315-0310-2. |

**Web Resources:**

| Sl. No. | Web Links |
|---|---|
| 1 | https://www.classcentral.com/subject/maths |
| 2 | https://academicearth.org/ |

**Course Articulation Matrix:**

| Course Outcomes | Program Outcomes | | | | | | | | | | | Program Specific Outcomes | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | P011 | PSO1 | PSO2 | PSO3 |
| CO1 | 3 | - | - | - | - | - | - | - | - | - | - | 2 | - | - |
| CO2 | 3 | - | - | - | - | - | - | - | - | - | - | 2 | - | - |
| CO3 | 3 | - | - | - | - | - | - | - | - | - | - | 2 | - | - |
| CO4 | 3 | - | - | - | - | - | - | - | - | - | - | 2 | - | - |
| CO5 | 3 | - | - | - | - | - | - | - | - | - | - | 2 | - | - |

High – 3, Medium – 2, Low – 1

| DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING | | | | | | |
|---|---|---|---|---|---|---|
| *Course Code:* 24MA313 | *Course Title:* ***Engineering Mathematics-III*** | | | | | |
| ***Total Credits: 03 L-45: T-0: P-0: SL and TW-45*** | ***SDGs addressed: 4, 5, 9, 16, 17*** | | ***Course Category: Basic Science Course*** | ***Course Type: Theory*** | | |
| ***Teaching Learning Process*** | | ***Total Contact Hours/semester*** | ***Assessment Weightage and Marks*** | | | |
| | ***Lecture*** | **45** | | **CIE** | **SEE** | **Total** |
| | ***Tutorial*** | **-** | ***Weightage %*** | ***40*** | ***60*** | ***100*** |
| | ***Practical*** | **-** | ***Maximum Marks*** | ***40*** | ***60*** | ***100*** |
| | ***Self-learning and Teamwork*** | **45** | ***Minimum Marks*** | ***20*** | ***25*** | ***45**** |
| | ***Total*** | **90** | | | | |

**Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)**

**Course Prerequisite:** Engineering Mathematics-I and Engineering Mathematics-II.

**Course Objectives:**

| Sl. No. | Course Objectives |
|---|---|
| 1 | To impart fundamental understanding of probability, statistics, Combinatorics and their applications in various Engineering fields. |

**Course Outcomes (COs)**

| CO | Course Outcomes | Highest Level of Cognitive Domain | Working Knowledge |
|---|---|---|---|
| CO1 | Apply least square method to fit a curve for the given data and evaluate the correlation coefficient and regression lines for the data. | L2 | WK1, WK3 |
| CO2 | Determine the nature of the events and hence calculate the appropriate probabilities of the events. | L2 | WK2, WK6 |
| CO3 | Classify the random variables to determine the appropriate probability distributions. | L2 | WK1, WK4 |
| CO4 | Apply the techniques of counting to identify the number of ways in which a given task can be accomplished without list all the possibilities explicitly. | L2 | WK7, WK8 |
| CO5 | Derive the generating function for the given situation and evaluate the required coefficient. | L2 | WK5, WK8 |

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

**Course Content / Syllabus:**

| Unit No. | Content | Lecture Hours |
|---|---|---|
| 1 | **Statistics:** Introduction, Definitions, Curve Fitting, equation of Straight line, parabola and exponential, correlation and regression, formula for correlation coefficient, regression lines and angle between the regression lines.<br>**Self-Learning and Teamwork***<br>• Least square method to derive normal equations.<br>• Fitting of curves: $y = ax^b$, $y = ab^x$.<br>• Find $r,\ \bar{x},\ \bar{y}$ when regression lines are given. | 09 |
| 2 | **Probability Theory:** Basic terminology, Definition of probability, Probability and set notations, Addition law of probability, independent events, conditional probability, multiplication law of probability, Baye's theorem.<br>**Self-Learning and Teamwork***<br>• Axioms of probability.<br>• Solve logical problems using the theory of probability.<br>• Solve real life problems using Bay's theorem. | 09 |
| 3 | **Random Variables:** Discrete Probability distribution, Continuous Probability distribution, expectation, Variance, Probability generating function, Binomial distribution, Poisson distribution, Normal distribution.<br>**Self-Learning and Teamwork**<br>• Plot Probability density functions for simple distributions.<br>• Visualize how binomial approximates Poisson as n→∞, p→0.<br>• Analyze how small changes in the PDF affect probabilities. | 09 |
| 4 | **Combinatorics-1:**<br>**Fundamental Principles of Counting:** The Rules of Sum and Product, Permutations (linear, circular, identical objects), Combinations – The Binomial Theorem, Multinomial Theorem and Combinations with Repetition.<br>**The Principle of Inclusion and Exclusion**: The Principle of Inclusion and Exclusion, Generalizations of the Principle.<br>**Self-Learning and Teamwork***<br>• Compare linear vs. circular arrangements with concrete examples.<br>• Compare outputs of $\binom{n}{r}$ and $P(n,r)$ for same values of n and r.<br>• Derive and compute small derangement numbers (e.g., for n = 3, 4, 5), Use PIE to solve problems like - how many numbers from 1 to 100 are not divisible by 2, 3, or 5? | 09 |

**Text Books:**

| Sl. No. | Author/s | Title | Publisher Details |
|---|---|---|---|
| 1 | B.S. Grewal | Higher Engineering Mathematics | 44th Edition, Khanna Publications, 2015. |
| 2 | Ralph P. Grimaldi | Discrete and Combinatorial Mathematics | 4th Edition, PHI/Pearson Education, 2005 |
| 3 | E. Kreyszig | Advanced Engineering Mathematics | 10th Edition (Reprint), 2016, John Wiley & Sons. |

**Reference Books:**

| Sl. No. | Author/s | Title | Publisher Details |
|---|---|---|---|
| 1 | N. P. Bali and Manish Goyal | A Textbook of Engineering Mathematics | Vol. I & II (AICTE), 10th Edition, Laxmi Publications Pvt Ltd, 2019. |
| 2 | Alan Tucker | Applied Combinatorics | 5th Edition, Wiley-India, 2011 |
| 3 | Maurice D. Weir, Joel R. Hass and George B. Thomas | Thomas' calculus: Early Transcendentals | 12th Edition, Pearson Education, 2016. |
| 4 | B. V. Ramana | Higher Engineering Mathematics | 11th Edition, Tata-McGraw Hill 2010. |
| 5 | Peter V.O 'Neil | Advanced Engineering Mathematics | 7th Edition, CENGAGE Learning India Pvt. Ltd. Publishers, 2012. |

**Web Resources:**

| Sl. No. | Web Links |
|---|---|
| 1 | https://www.classcentral.com/subject/maths |
| 2 | https://academicearth.org/ |

**Course Articulation Matrix:**

| Course Outcomes | Program Outcomes | | | | | | | | | | | Program Specific Outcomes | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 | PSO3 |
| CO1 | 3 | - | - | - | - | - | - | - | - | - | - | 2 | - | - |
| CO2 | 3 | - | - | - | - | - | - | - | - | - | - | 2 | - | - |
| CO3 | 3 | - | - | - | - | - | - | - | - | - | - | 2 | - | - |
| CO4 | 3 | 1 | - | - | - | - | - | - | - | - | - | 2 | - | - |
| CO5 | 3 | 1 | - | - | - | - | - | - | - | - | - | 2 | - | - |

High – 3, Medium – 2, Low – 1

| DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING | | | | | | |
|---|---|---|---|---|---|---|
| *Course Code:* 24CS310 | Course Title: *Object Oriented Programming using C++* | | | | | |
| *Total Credits: 04 L-45: T-0: P-30: SL and TW-45* | *SDGs addressed: 4, 8, 9, 12,13* | *Course Category: PCC* | *Course Type: Theory with Practical* | | | |
| *Teaching Learning Process* | | *Total Contact Hours/Semester* | *Assessment in Weightage and marks* | | | |
| | *Lecture* | 45 | | CIE | SEE | Total |
| | *Tutorial* | - | *Weightage in %* | *40* | *60* | *100* |
| | *Practical* | 30 | *Maximum Marks* | *40* | *60* | *100* |
| | *Self-learning and Term Work* | 45 | *Minimum Marks* | *20* | *25* | *45** |
| | *Total* | 120 | | | | |

**Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)**

**Course Prerequisite:** C programming concepts.

**Course Objectives:**

| Sl. No. | Course Objectives |
|---|---|
| 1 | Understand the object-oriented programming paradigm and Gain knowledge about how information is stored in an object. |
| 2 | Understand the capability of a class to rely upon another class and functions. |
| 3 | Understand about constructors and their uses. |
| 4 | Create and process data in files using file I/O functions. |
| 5 | Use the generic programming features of C++ including Exception handling. |

**Course Outcomes (COs)**

| CO | Course Outcomes | Highest Level of Cognitive Domain | Working Knowledge |
|---|---|---|---|
| CO1 | Understand object-oriented programming concepts and be able to design solutions for a given problem. | L2 | WK2, WK4 |
| CO2 | Able to reuse code with extensible Class types, user-defined operators and function overloading. | L2 | WK3, WK4, WK6 |
| CO3 | Achieve code reusability and extensibility by means of Inheritance and Polymorphism | L3 | WK3, WK4, WK6, WK8 |
| CO4 | Incorporate file-handling functions and exceptions for a given application. | L3 | WK3, WK4, WK6, WK7 |

| | | | |
|---|---|---|---|
| CO5 | Design C++ packages with templates for providing solutions to complex problems. | L3 | WK4, WK6, WK8 |

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

**Course Content / Syllabus:**

| UNIT No. | Content | Hours Lecture |
|---|---|---|
| 1 | **Principles of Object-Oriented Programming:** Software crisis, software evolution, A look at procedure-oriented programming, object-oriented programming paradigm, basic concepts of object-oriented programming, benefits of OOP, object-oriented languages, applications of OOP.<br>**Beginning with C++:** What is C++?; application of C++, a simple C++ program, more C++ statements, an example with Class, structure of C++ program, creating the source file, compiling and linking.<br>**Tokens, Expressions and Control Structures:** Introduction, tokens, keywords, identifiers and constants, basic datatypes, user-defined datatypes, storage classes, derived datatypes, symbolic constants, type compatibility, declaration of variables, dynamic initialization of variables, reference variables, operators in C++, scope resolution operator, member dereferencing operators, memory management operators, manipulators, type cast operators, expressions and their types, special assignment expressions, implicit conversions, operator overloading, operator precedence, control structures.<br>**Functions in C++:** Introduction, the main() function, function prototyping, call by reference, return by reference, inline functions, default arguments, 'const' arguments, recursion, function overloading, friend and virtual functions, Math library functions.<br>**Self-Learning and Term Work***<br>• Explore the historical evolution from procedural languages (C, Pascal) to OOP and compare with modern paradigms (Python, Java).<br>• Investigate the role of OOP in real-world applications such as GUI frameworks or game development. | 09 |
| 2 | **Classes and Objects:** Introduction, C structure revisited, specify a class, defining member functions, C++ program with Class, making an outside function inline, nesting of member functions, private member functions, arrays within a class, memory allocation for objects, static data members, static member functions, arrays of objects, object as function arguments, friendly functions, returning objects, 'const' members functions, pointers to members.<br>**Constructors and Destructors:** Introduction, constructors, parameterized constructors, multiple constructors in a class, constructor with default arguments, dynamic initialization of objects, copy constructor, dynamic constructors, constructing two-dimensional arrays, destructors.<br>**Operator Overloading and Type Conversions:** Introduction, defining operator overloading, overloading unary and binary operators, | 09 |

| | | |
|---|---|---|
| | overloading binary operator using friend, manipulation of string using operators, some other operators overloading, rules for overloading operators, type conversion.<br>**Self-Learning and Term Work***<br>• Explore memory management strategies: stack vs heap allocation in C++ compared to other languages.<br>• Study smart pointers (e.g., unique_ptr, shared_ptr) as an extension to memory management in classes. | |
| 3 | **Inheritance: Extending Classes** Introduction, defining derived classes, single inheritance, making a private members inheritable, multilevel inheritance, multiple inheritance, hierarchical inheritance, virtual base classes, abstract classes, constructors in derived classes, member classes: nesting of classes.<br>**Pointers, Virtual Functions and Polymorphism:** Introduction, pointers, pointers to objects, 'this' pointer, polymorphism, pointers to derived classes, virtual functions, pure virtual functions, virtual constructors and destructors.<br>**Self-Learning and Term Work***<br>• Study the difference between early binding (static polymorphism) and late binding (dynamic polymorphism) and write a simple program using both function overloading and virtual functions to demonstrate each concept.<br>• Explore polymorphism in standard libraries (STL iterators, streams). | 09 |
| 4 | **Managing Console I/O Operations:** introduction, C++ streams, C++ stream classes, unformatted I/O operations, formatted console I/O operations, managing output with manipulators.<br>**Working with Files:** Introduction, classes for file stream operations, opening and closing files, detecting end-of-file.<br>**Exception Handling:** Introduction, basics of exception handling, exception handling mechanism, throwing mechanism, catching mechanism.<br>**Self-Learning and Term Work***<br>• Develop a console-based C++ application (e.g., a simple text processor), dividing tasks for I/O operations and exception handling.<br>• Custom exception classes in C++ | 09 |
| 5 | **Templates:** Introduction, class templates, class templates with multiple parameters, function templates, function templates with multiple parameters, overloading of template functions, member function templates.<br>**Self-Learning and Term Work***<br>• Explore how templates enable generic programming and influence modern C++ libraries.<br>• Implement a generic swap function using a function template and demonstrate how this single function can be used to swap variables of different data types (e.g., int, double, string) without needing to write multiple functions. | 09 |

*** NOTE: Self-Learning and Term Work are evaluated only through continuous internal evaluation (CIE) and will not be assessed during the Semester End Examination (SEE).**

**Text Books:**

| Sl. No. | Author/s | Title | Publisher Details |
|---|---|---|---|
| 1 | Herbert Schildt | C++: The Complete Reference | 12th Edition, Tata McGraw Hill Education, 5 December 2022, ISBN-13: 978-9355323712 |
| 2 | E Balagurusamy | Object-Oriented Programming with C++ | 8th Edition, Tata McGraw Hill Education, 24 September 2020, ISBN-13: 978-9389949186 |

**Reference Books:**

| Sl. No. | Author/s | Title | Publisher Details |
|---|---|---|---|
| 1 | Bjarne Stroustrup | C++ PROGRAMMING LANGUAGE | 4th Edition, Pearson Education, 31 May 2022, ISBN-13: 978-9356060135 |
| 2 | Stanley B.Lippmann, Josee Lajore | C++ Primer | 4th Edition, Pearson Education, 2005. |
| 3 | Yashavant Kanetkar | Let Us C++ | BPB Publications Edition, 16 September 2020, ISBN-13: 978-9388176644 |
| 4 | Behrouz A. Forouzan, Richard F. Gilberg | C++ Programming: An Object-Oriented Approach | First Edition, Tata McGraw Hill Education, 19 May 2022, ISBN-13: 978-9355321305 |

**Web Resources:**

| Sl. No. | Web Links |
|---|---|
| 1 | Programming in C++: https://archive.nptel.ac.in/courses/106/105/106105151/ |
| 2 | Programming In Modern C++: https://onlinecourses.nptel.ac.in/noc22_cs103/preview |

**Course Articulation Matrix:**

| Course Outcomes | Program Outcomes | | | | | | | | | | | Program Specific Outcomes | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 | PSO3 |
| CO1 | 3 | - | - | - | - | - | - | - | - | - | - | - | - | - |
| CO2 | 3 | 3 | 3 | - | - | - | - | 3 | 3 | - | - | 3 | - | 3 |
| CO3 | 3 | 3 | 3 | - | 3 | - | - | 3 | 3 | - | - | 3 | - | 3 |
| CO4 | 3 | 3 | 3 | 1 | 3 | - | - | 3 | 3 | - | - | - | - | 3 |
| CO5 | 3 | 3 | 3 | - | 3 | - | - | - | - | - | - | 3 | - | 3 |

High – 3, Medium – 2, Low – 1

**LIST OF PROGRAMS FOR PRACTICAL SESSIONS**

| Weeks | List of Programs | No. of Hours |
|---|---|---|
| 1 | a) Write a C++ program to find and output all the roots of a quadratic equation.<br>b) Write a C++ program to find the sum of all the natural numbers from 1 to n.<br>c) Write a C++ program to sort the elements in ascending and descending order. | 02 |
| 2 | The daily maximum temperatures recorded for 5 cities during the first 6 days of January month have to be tabulated. Write a C++ program to read the data and find the city and day corresponding to highest temperature and lowest temperature. | 02 |
| 3 | a)Write a C++ program to swap 2 values by writing functions for the following:<br>i. Call-by-name/value<br>ii. Call-by-pointers and<br>iii. Call-by-reference<br>b) Define a Cube class and write a C++ program to read the side for three cubes and print their volume and outer area. | 02 |
| 4 | a)An election is contested by 5 candidates. The candidates are numbered 1 to 5 and the voting is done by marking the candidate number on the ballot paper. Develop an application to read the ballots and count the votes cast for each candidate using an array variable count. In case, a number read is outside the range 1 to 5, the ballot should be considered as a 'spoilt ballot' and the program should also count the number of spoilt ballots.<br>b) Write a C++ program to demonstrate function overloading for the following prototypes.<br>sum_two_nums(int a, int b);<br>sum_two_nums(double a, double b); | 02 |
| 5 | Given that an EMPLOYEE class contains following members:<br>Data members: Employee_Number, Employee_Name, Basic, DA, IT, Net_Salary.<br>Member functions: to read the data, to calculate Net_Salary and to print data members.<br>Write a C++ program to read the data of N employees and compute Net_Salary of each employee.<br>(Dearness Allowance (DA) = 52% of Basic and Income Tax (IT) = 30% of the gross salary. Net_Salary = Basic + DA - IT) | 02 |
| 6 | Define a STUDENT class with USN, Name, and Marks in 3 tests of a subject. Declare an array of 10 STUDENT objects. Using appropriate functions, find the average of two better marks for each student. Print the USN, Name, and the average marks of all the students. | 02 |
| 7 | Write a C++ program to create a class called COMPLEX and implement the following overloading functions ADD that return a | 02 |

| | | |
|---|---|---|
| | COMPLEX number.<br>i. ADD (a, s2) – where a is an integer (real part) and s2 is a complex number.<br>ii. ADD (s1, s2) – where s1 and s2 are complex numbers. | |
| 8 | Create a class named Shape with a function that prints "This is a shape". Create another class named Polygon inheriting the Shape class with the same function that prints "Polygon is a shape". Create two other classes named Rectangle and Triangle having the same function which prints "Rectangle is a polygon" and "Triangle is a polygon" respectively. Again, make another class named Square having the same function which prints "Square is a rectangle". Now, try calling the function by the object of each of these classes. Write a C++ program to demonstrate the same. | 02 |
| 9 | Suppose we have three classes Vehicle, FourWheeler, and Car. The class Vehicle is the base class, the class FourWheeler is derived from it and the class Car is derived from the class FourWheeler. Class Vehicle has a method 'vehicle' that prints 'I am a vehicle', class FourWheeler has a method 'fourWheeler' that prints 'I have four wheels', and class Car has a method 'car' that prints 'I am a car'. So, as this is a multi-level inheritance; we can have access to all the other classes methods from the object of the class Car. We invoke all the methods from a Car object and print the corresponding outputs of the methods. So, if we invoke the methods in this order, car(), fourWheeler(), and vehicle(), then the output will be,<br>• I am a car<br>• I have four wheels<br>• I am a vehicle<br>Write a C++ program to demonstrate multilevel inheritance using the above. | 02 |
| 10 | Write a function which throws a division by zero exception and catch it in catch block. Write a C++ program to demonstrate usage of try, catch and throw to handle exception. | 02 |
| 11 | Write a C++ program function which handles array of bounds exception using C++. | 02 |
| 12 | Write a C++ program that consists of Stack class and include methods to push and pop the stack elements. Create your own defined Exception class and handle exceptions while stacks run with “Stack as full” and “Stack under flow” while calling push() and pop() methods respectively. Implement this<br>using ‘throws’ and ‘throw’ keywords. | 02 |
| 13 | **Revision** | 02 |
| 14 | **Revision** | 02 |
| 15 | **Laboratory Test/Event** | 02 |

| DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING | | | | | | |
|---|---|---|---|---|---|---|
| ***Course Code:*** **24CS320** | | **Course Title:** ***Digital System Design*** | | | | |
| ***Total Credits: 04***<br>***L-45: T-0: P-30: SL and TW-45*** | | ***SDGs addressed: 4, 8, 9*** | ***Course Category: PCC*** | | ***Course Type: Theory with Practical*** | |
| ***Teaching Learning Process*** | | ***Total Contact Hours/ Semester*** | ***Assessment in Weightage and marks*** | | | |
| | ***Lecture*** | **45** | | **CIE** | **SEE** | **Total** |
| | ***Tutorial*** | **-** | ***Weightage in %*** | ***40*** | ***60*** | ***100*** |
| | ***Practical*** | **30** | ***Maximum Marks*** | ***40*** | ***60*** | ***100*** |
| | ***Self-learning and Term Work*** | **45** | ***Minimum Marks*** | ***20*** | ***25*** | ***45**** |
| | ***Total*** | **120** | | | | |

**Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)**

**Course Prerequisite:** NIL

**Course Objectives:**

| Sl. No. | Course Objectives |
|---|---|
| 1 | Understand Boolean algebra and basic properties of Boolean algebra; able to simplify simple Boolean functions by using the basic Boolean properties. |
| 2 | Able to design simple combinational logics using basic gates. Able to optimize simple logic using Karnaugh maps, understand "don't care". |
| 3 | Familiar with basic sequential logic components: SR Latch, D Flip-Flop and their usage and able to analyze sequential logic circuits. |
| 4 | Understand finite state machines (FSM) concepts. |

**Course Outcomes (COs)**

| CO | Course Outcomes | Highest Level of Cognitive Domain | Working Knowledge |
|---|---|---|---|
| CO1 | Explain the fundamental concepts of Boolean algebra, Boolean theorems, K-MAP, Quine-McCluskey algorithm and apply the knowledge of theoretical foundations to simplify and solve Boolean functions. | L3 | WK1, WK2 |
| CO2 | Design and implement the combinational circuits. | L3 | WK1, WK2 |
| CO3 | Apply the knowledge of timing aspects and operation of Flip-Flops in conversion between different Flip Flops. | L3 | WK1, WK3 |
| CO4 | Illustrate the working of various registers and register circuits. | L2 | WK1, WK3 |
| CO5 | Design and implement synchronous/asynchronous counters and synchronous sequential circuits. | L3 | WK1, WK6 |

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

**Course Content / Syllabus:**

| UNIT No. | Content | Hours Lecture |
|---|---|---|
| 1 | **Digital Logic:** Combinational Logic Circuits, Boolean laws and Theorems, Sum-of-Products Method, Truth Table to Karnaugh Map, Pairs, Quads, and Octets, Karnaugh Simplifications, Don't-care Conditions, Product-of-sums Method, Product-of-sums simplifications, Simplification by Quine- McCuskey Method.<br>**Self-Learning and Term Work***<br>• Number System<br>• Axiomatic Definition of Boolean Algebra or Huntington Postulates | 09 |
| 2 | **Data Processing Circuits:** Multiplexers, Demultiplexers, 1-of-16 Decoder, BCD to Decimal Decoders, Encoders, Parity Generators and Checkers, Magnitude Comparator.<br>**Self-Learning and Term Work***<br>• Parallel adder with look-ahead method<br>• Conversion from one combinational circuit to another. | 09 |
| 3 | **Flip-Flops**: RS Flip-Flops, Gated Flip-Flops, Edge-triggered RS FLIP-FLOP, Edge-triggered D FLIP-FLOPs, Edge-triggered JK FLIP-FLOPs, JK Master-slave FLIP-FLOP, Switch Contact Bounce Circuits, Various Representation of FLIP-FLOPs, Conversion of Flip-Flops.<br>**Self-Learning and Term Work***<br>• Designing of different flip-flops using state tables and state diagrams.<br>• Flip-flops and Counter design. | 09 |
| 4 | **Registers:** Types of Registers, Serial In - Serial Out, Serial In - Parallel out, Parallel In - Serial Out, Parallel In - Parallel Out, Universal Shift Register, Applications of Shift Registers.<br>**Self-Learning and Term Work***<br>• Design process of registers, clock synchronization, bit shifting, and resetting mechanisms | 09 |
| 5 | **Counters:** Asynchronous Counters, Synchronous Counters, Decade Counters. **Design of Synchronous Sequential Circuits**: Model Selection, State Transition Diagram, State Synthesis table, Design equations and Circuit diagram.<br>**Self-Learning and Term Work***<br>• Design of Mod-N counters.<br>• Study of Basic Moore and Mealy Machine. | 09 |

*** NOTE: Self-Learning and Term Work are evaluated only through continuous internal evaluation (CIE) and will not be assessed during the Semester End Examination (SEE).**

**Text Book:**

| Sl. No. | Author/s | Title | Publisher Details |
|---|---|---|---|
| 1 | Donald P Leach, Albert Paul Malvino and Goutam | Digital Principles and Applications | 7th Edition, Tata McGraw Hill, 2014. |

| | | | |
|---|---|---|---|
| | Saha | | |

**Reference Books:**

| Sl. No. | Author/s | Title | Publisher Details |
|---|---|---|---|
| 1 | Morris Mano, Kim and Charles | Logic and Computer Design Fundamentals | 5th Edition, Prentice Hall, 2016. |
| 2 | Donald D.Givone | Digital Principles and Design | 1st Edition, Tata McGraw-Hill, 2012. |
| 3 | John M.Yarbrough | Digital Logic Applications and Design | 1st Edition, Thomson Learning, 2001 |
| 4 | Kohavi | Switching and Finite Automata Theory | 3rd Edition, TMH, 2010. |

**Web Resources:**

| Sl. No. | Web Links |
|---|---|
| 1 | https://nptel.ac.in/courses/108105113/15 |
| 2 | https://nptel.ac.in/courses/117105080/19 |

**Course Articulation Matrix:**

| Course Outcomes | Program Outcomes | | | | | | | | | | | Program Specific Outcomes | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 | PSO3 |
| CO1 | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 3 | - | - |
| CO2 | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 3 | 3 | 3 |
| CO3 | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 3 | 3 | 3 |
| CO4 | 3 | 3 | - | - | - | - | - | - | - | - | - | 3 | 3 | 3 |
| CO5 | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 3 | 3 | - |

High – 3, Medium – 2, Low – 1

**LIST OF PROGRAMS FOR PRACTICAL SESSIONS**

| Weeks | List of Programs | No. of Hours |
|---|---|---|
| 1 | a. Study and verify the truth table of logic gates.<br>b. Realize Basic Gates using Universal Gates.<br>c. Simplify the given expression and realize it using gates. | 02 |
| 2 | Design and realize<br>a. Half Adder / Full Adder.<br>b. Half Subtractor / Full Subtractor. | 02 |
| 3 | Design and set up the following circuit using IC 7483.<br>a. A 4-bit binary parallel adder.<br>b. A 4-bit binary parallel subtractor. | 02 |
| 4 | Design and realize the following using Basic gates & IC 7483.<br>a. BCD to Excess- 3Code and vice versa<br>b. Binary to Gray Code and vice versa | 02 |

| | | |
|---|---|---|
| 5 | Design and implement<br>a. 4:1 Multiplexer (MUX) using only NAND gates.<br>b. Boolean expressions using 8:1 Multiplexer(MUX). | 02 |
| 6 | Half/Full Adder and Half/Full Subtractor using IC 74153. | 02 |
| 7 | Design and implement One & Two-Bit Magnitude Comparator. | 02 |
| 8 | Design and implement<br>a. Decoder circuit using basic gates and to verify using IC74LS138.<br>b. Half/Full Adder and Half/Full Subtractor using IC74LS138. | 02 |
| 9 | a. Design and implement 3-bit parity generator and verify using Parity Checker.<br>b. Implementation and verification of truth table for J-K flip-flop, Master-slave J- K flip-flop, D flip-flop and T flip-flop. | 02 |
| 10 | Design and implementation of Mod-N synchronous counter using J-K flip-flops. | 02 |
| 11 | Design and implementation of shift register to function as i) SISO ii) SIPO iii) PISO iv) PIPO v) shift left and vi) shift right operation. | 02 |
| 12 | a. Design and implementation of i) Ring counter and ii) Johnson counter using 4- bit shift register.<br>b. Design and implementation of Decade Counters<br>c. Design and implementation of Asynchronous Counters | 02 |
| 13 | **Revision** | 02 |
| 14 | **Revision** | 02 |
| 15 | **Laboratory Test/Event** | 02 |

| DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING | | | | | | |
|---|---|---|---|---|---|---|
| ***Course Code:*** **24CS330** | ***Course Title*** ***Data Structures*** | | | | | |
| ***Total Credits: 04***<br>***L-60: T-0: P-0: SL and TW-60*** | ***SDGs addressed: 4, 9, 11*** | | ***Course Category: PCC*** | ***Course Type: Theory*** | | |
| ***Teaching Learning Process*** | | ***Total Contact Hours/semester*** | ***Assessment Weightage and Marks*** | | | |
| | ***Lecture*** | **60** | | **CIE** | **SEE** | **Total** |
| | ***Tutorial*** | **-** | ***Weightage, %*** | ***40*** | ***60*** | ***100*** |
| | ***Practical*** | **-** | ***Maximum Marks*** | ***40*** | ***60*** | ***100*** |
| | ***Self-learning and Term Work*** | **60** | ***Minimum Marks*** | ***20*** | ***25*** | ***45**** |
| | ***Total*** | **120** | | | | |

**Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)**

**Course Prerequisite:** C programming concepts.

**Course Objectives:**

| Sl. No. | Course Objectives |
|---|---|
| 1 | Provide the knowledge of basic data structures and their implementations. |
| 2 | Introduce how arrays, linked structures, stacks, queues, trees, and graphs are represented in memory and used by algorithms in problem solving. |
| 3 | Impart the benefits of dynamic and static data structures. |
| 4 | Understand the importance of data structures in the context of writing efficient programs. |

**Course Outcomes (COs)**

| CO | Course Outcomes | Highest Level of Cognitive Domain | Working Knowledge |
|---|---|---|---|
| CO1 | Using the concept of pointers to arrays, structures, and functions, perform different operations on linear list with array storage representation. | L3 | WK2,WK3 |
| CO2 | Use the concept of linked list representation to implement various operations on Linear lists with different types of linked list. | L3 | WK2,WK4 |
| CO3 | Solve the problems of conversion and evaluation of different types of expressions using stack data structure, comprehend the implementation of different types of Queues. | L3 | WK2,WK4 |
| CO4 | Using tree data structure solve the problems of searching, sorting, expression conversion and expression evaluation. | L3 | WK2,WK4 |
| CO5 | Applying graph traversal techniques solve the | L3 | WK2,WK4 |

| | | | |
|---|---|---|---|
| | problems of testing connectivity and acyclicity in a graph. | | |

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

**Course Content / Syllabus:**

| Unit No. | Content | Lecture Hours |
|---|---|---|
| 1 | **Introduction to Data structures:** Basic Terminology, Elementary Data Structure Organization, Classification of Data Structures, Operations on Data Structures, Abstract Data Type.<br>**Pointers:** Declaring Pointer Variables, Pointer Expressions and Pointer Arithmetic Null Pointers, Generic Pointers, Pointer to Pointers, Drawback of Pointers.<br>**Arrays:** Pointers and Arrays, Arrays of Pointers, Pointers and Two-dimensional Arrays.<br>**Structures and Unions:** Introduction, Nested Structures, Arrays of Structures, Structures and Functions, Self-referential Structures.<br>**Self-Learning and Term Work***<br>• Multi-dimensional Arrays.<br>• Strings.<br>• Unions | 12 |
| 2 | **Linked Lists:** Introduction, Singly Linked Lists, Circular Linked Lists, Doubly Linked Lists, Circular Doubly Linked Lists, Header Linked Lists<br>**Self-Learning and Term Work***<br>• Multi-linked Lists.<br>• Applications of Linked Lists. | 12 |
| 3 | **Stacks:** Introduction to Stacks, Array Representation of Stacks , Operations on a Stack, Linked Representation of Stacks, Operations on a Linked Stack,<br>Applications of Stacks,<br>**Queues:** Introduction to Queues, Array Representation of Queues, Linked Representation of Queues ,Types of Queues<br>**Self-Learning and Term Work***<br>• Multiple Stacks<br>• Multiple Queues.<br>• Applications of Queues | 12 |
| 4 | **Trees:** Introduction, Types of Trees, Creating a Binary Tree from a General Tree, Traversing a Binary Tree<br>**Efficient Binary Trees:** Binary Search Trees, Operations on Binary Search Tree, Binary Expression Trees, Threaded Binary Trees<br>**Self-Learning and Term Work***<br>• Applications of Trees. | 12 |

| | | |
|---|---|---|
| 5 | **Graphs:** Introduction, Graph Terminology, Directed Graphs, Bi-connected Components, Representation of Graphs, Graph Traversal Algorithms.<br>**Self-Learning and Term Work***<br>• Files and Their Organization | 12 |

*** NOTE: Self-Learning and Term Work are evaluated only through continuous internal evaluation (CIE) and will not be assessed during the Semester End Examination (SEE).**

**Text Books:**

| Sl. No | Author/s | Title | Publisher Details |
|---|---|---|---|
| 1 | E Balaguruswamy | Programming in ANSI C | 8th Edition, McGraw Hill Education, 2019. |
| 2 | Richard F.Gilberg and Behronz A. Forouzan | Data structures, A Pseudo code Approach with C | 3rd Edition, Cengage learning, 2013. |

**Reference Books:**

| Sl. No. | Author/s | Title | Publisher Details |
|---|---|---|---|
| 1 | Yedidyah, Augenstein, Tannenbaum | Data Structures Using C and C++ | 2nd Edition, PHI. |
| 2 | E Balaguruswamy | Data Structures Using C | McGraw Hill Education, Paperback 2017. |
| 3 | Ellis Horowitz and Sartaj Sahni | Fundamentals of Data Structures in C | 2nd edition, Universities Press, 2014. |
| 4 | Seymour Lipschutz | Data Structures Schaum's Outlines | Revised 1st Edition, McGraw Hill, 2014. |

**Web Resources:**

| Sl. No. | Web Links |
|---|---|
| 1 | https://nptel.ac.in/courses/106102064/ |
| 2 | https://nptel.ac.in/courses/106105085/ |

**Course Articulation Matrix:**

| Course Outcomes | Program Outcomes | | | | | | | | | | | Program Specific Outcomes | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 | PSO3 |
| CO1 | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 3 | - | 3 |
| CO2 | 3 | 3 | 3 | 3 | - | - | - | - | - | - | - | 3 | - | 3 |
| CO3 | 3 | 3 | 3 | 3 | - | - | - | - | - | - | - | 3 | - | 3 |
| CO4 | 3 | 3 | 3 | 3 | - | - | - | - | - | - | - | 3 | - | 3 |
| CO5 | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 3 | - | 3 |

High – 3, Medium – 2, Low – 1

| DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING | | | | | | |
|---|---|---|---|---|---|---|
| *Course Code: 24CS340* | *Course Title: Computer Organization* | | | | | |
| *Total Credits: 03*<br>*L-45: T-0: P-0: SL and TW-45* | | *SDGs addressed: 4,9* | *Course Category: PCC* | *Course Type: Theory* | | |
| *Teaching Learning Process* | | *Total Contact Hours/semester* | *Assessment Weightage and Marks* | | | |
| | *Lecture* | 45 | | CIE | SEE | Total |
| | *Tutorial* | - | *Weightage, %* | *40* | *60* | *100* |
| | *Practical* | - | *Maximum Marks* | *40* | *60* | *100* |
| | *Self-learning and Term Work* | 45 | *Minimum Marks* | *20* | *25* | *45** |
| | *Total* | 90 | | | | |

**Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)**

**Course Prerequisite: NIL**

**Course Objectives:**

| Sl. No. | Course Objectives |
|---|---|
| 1 | Understand the structure, function and characteristics of computer system. |
| 2 | Identify the elements of modern instructions sets and their impact on processor design. |
| 3 | Learn different ways of communicating with I/O devices and hierarchical memory systems like cache memory and virtual memory. |
| 4 | Describe ALU operations with integer and floating–point operands and understand the basic processing unit and organization of simple processor and other large computing systems. |

**Course Outcomes (COs)**

| CO | Course Outcomes | Highest Level of Cognitive Domain | Working Knowledge |
|---|---|---|---|
| CO1 | Apply the basic concepts of computer, instruction set architecture, function and characteristics of a computer system and design the assembly language for the sequence of machine instruction. | L3 | WK2, WK3 |
| CO2 | Explain the different ways of communicating with the I/O devices. | L2 | WK3, WK4 |
| CO3 | Design and analyse the performance of memory systems. | L3 | WK2, WK3 |
| CO4 | Apply the Arithmetic and Logical operations with integer and floating–point operands. | L3 | WK2, WK3 |
| CO5 | Illustrate the basic processing unit and organization of simple processor. | L2 | WK3, WK4 |

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

**Course Content / Syllabus:**

| UNIT No. | Content | Hours Lecture |
|---|---|---|
| 1 | **Basic Structures of Computers, Machine Instructions & Programs:** Operational Concepts, Number Representations and Arithmetic Operations, Character Representation, Performance. **Instruction Set Architecture:** Memory Locations and Addresses, Memory Operations, Instructions and Instruction Sequencing, Addressing Modes, Assembly Language, Stacks, Subroutines, Additional Instructions, CISC Instruction Sets, RISC and CISC styles.<br>**Self-Learning and Term Work***<br>• The Stack frames for nested subroutines<br>• Additional Instructions: Multiplication and Division | 09 |
| 2 | **Input/Output Organization:** Basic Input/Output, Accessing I/O Devices, Interrupts, Bus Structure, Bus Operation, Arbitration.<br>**Self-Learning and Term Work***<br>• An example of RISC –Style I/O Program<br>• An example of CISC –Style I/O Program<br>• Examples of Interrupt Programs | 09 |
| 3 | **The Memory System:** Basic Concepts, Semiconductor RAM Memories, Read Only Memories, Direct Memory Access, Memory Hierarchy, Cache Memories, Performance Considerations, Virtual Memory.<br>**Self-Learning and Term Work***<br>• Examples of Mapping Techniques<br>• Performance Considerations | 09 |
| 4 | **Arithmetic Operations:** Addition and Subtraction of Signed Numbers, Design of Fast Adders, Multiplication of Signed Numbers, Fast Multiplication: Bit-Pair Recoding of Multipliers, Integer Division, Floating-point Numbers and Operations.<br>**Self-Learning and Term Work***<br>• Examples of Floating-point operations | 09 |
| 5 | **Basic Processing Unit:** Some Fundamental Concepts, Instruction Execution, Hardware Components: Register File, ALU, Datapath, Hardwired Control: Generation of Control Signals, CISC style Processors<br>**Self-Learning and Term Work***<br>• CISC style Processors: An interconnect using buses | 09 |

**NOTE: *Self-Learning and Term Work are evaluated only through continuous internal evaluation (CIE) and will not be assessed during the Semester End Examination (SEE).**

**Text Book:**

| Sl. No. | Author/s | Title | Publisher Details |
|---|---|---|---|

| | | | |
|---|---|---|---|
| 1 | Carl V Hamacher, Zvonko Vranesic | Computer Organization and Embedded Systems | 6th Edition, McGraw Hill, 2012 |

**Reference Books:**

| Sl. No. | Author/s | Title | Publisher Details |
|---|---|---|---|
| 1 | M. MorisMano | Computer Systems Architecture | 3rd Edition, Pearson/PHI, 2007. |
| 2 | Carl Hamacher, Zvonko Vranesic, Safwat Zaky | Computer Organization | 5th Edition, Tata McGraw Hill, 2002. |
| 3 | William Stallings | Computer Organization & Architecture | 9th Edition, Pearson, 2015 |
| 4 | V Heuring and H Jordan | Computer System Design &Architecture | 1st Edition, Addison-Wesley, 1999. |

**Web Resources:**

| Sl. No. | Web Links |
|---|---|
| 1 | https://nptel.ac.in/courses/106106092/ |
| 2 | https://www.youtube.com/watch?v=CDO28Esqmcg |

**Course Articulation Matrix:**

| Course Outcomes | Program Outcomes | | | | | | | | | | | Program Specific Outcomes | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 | PSO3 |
| CO1 | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 3 | 3 | 3 |
| CO2 | 3 | 3 | - | - | - | - | - | - | - | - | - | 3 | 3 | - |
| CO3 | 3 | 3 | - | - | - | - | - | - | - | - | - | 3 | 3 | 3 |
| CO4 | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 3 | 3 | 3 |
| CO5 | 3 | 3 | 3 | 3 | - | - | - | - | - | - | - | 3 | 3 | - |

High – 3, Medium – 2, Low – 1

| DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING | | | | | | |
|---|---|---|---|---|---|---|
| *Course Code: 24CS350* | *Course Title: Discrete Mathematical Structures* | | | | | |
| *Total Credits: 03*<br>**L-45: T-0: SL and TW-45** | *SDGs addressed: 4, 9, 11, 16,17* | | *Course Category: PCC* | *Course Type: Theory* | | |
| *Teaching Learning Process* | | *Total Contact Hours/semester* | *Assessment Weightage and Marks* | | | |
| | *Lecture* | **45** | | **CIE** | **SEE** | **Total** |
| | *Tutorial* | - | *Weightage, %* | *40* | *60* | *100* |
| | *Practical* | - | *Maximum Marks* | *40* | *60* | *100* |
| | *Self-learning and Term Work* | **45** | *Minimum Marks* | *20* | *25* | *45** |
| | *Total* | **90** | | | | |

**Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)**

**Course Prerequisite:** Basic mathematics.

**Course Objectives:**

| Sl. No. | Course Objectives |
|---|---|
| 1 | Introduce concepts of mathematical logic for analyzing propositions and proving theorems. |
| 2 | Work with relations, digraphs and investigate their properties. |
| 3 | Introduce the concepts of coding, decoding theory, trees and graphs. |

**Course Outcomes (COs)**

| CO | Course Outcomes | Highest Level of Cognitive Domain | Working Knowledge |
|---|---|---|---|
| CO1 | Use the knowledge needed to test the logic of the program. | L3 | WK2, WK3 |
| CO2 | Demonstrate the concept of relations and be able to determine their properties. | L3 | WK2, WK3 |
| CO3 | Explain and apply decode group codes based on the various methods. | L3 | WK2, WK4 |
| CO4 | Identify different types of graphs such as Isomorphism, Bipartite, planar, Hamiltonian paths and cycles, Euler trails and circuits based on their properties. | L3 | WK2, WK4 |
| CO5 | Explain different types of trees and apply tree traversal techniques like preorder and postorder. | L3 | WK2, WK3 |

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

**Course Content / Syllabus:**

| UNIT No. | Content | Hours Lecture |
|---|---|---|
| 1 | **Fundamental of Logic:** Basic Connectives and Truth Tables, Logical Equivalence: The Laws of Logic, Logical Implication: Rules of Inference, The Use of Quantifiers, Quantifiers.<br>**Self-Learning and Team Work*:**<br>• Practice creating and verifying truth tables independently.<br>• Explore real-life applications of logical equivalence (digital circuits, program verification).<br>• Peer evaluation of logical arguments to strengthen reasoning. | 09 |
| 2 | **Relations:** Cartesian Products and Relations, Properties of relations, Computer Recognition: Zeros- One Matrices and Directed Graphs, Partial; orders: Hasse diagrams, Equivalence Relations and Partitions, Lattices.<br>**Self-Learning and Team Work*:**<br>• Study and classify relations using examples from databases or social networks.<br>• Explore lattice applications in decision-making systems individually. | 09 |
| 3 | **Coding Theory:** Elements of coding theory and Hamming Metric, generation of codes using Parity check and Generator matrices.<br>**Self-Learning and Team Work*:**<br>• Learn to decode and encode messages using Hamming codes or generator matrices through individual assignments.<br>• Implement group coding exercises to simulate error detection and correction in a communication channel. | 09 |
| 4 | **Graph theory:** Definitions and Examples, Subgraphs, Complements, and Graph Isomorphism, Vertex degree, Euler trail and circuits, Planar Graphs, Hamiltonian Paths and cycles.<br>**Self-Learning and Team Work*:**<br>• Independently identify graph types and verify properties using small examples.<br>• Use software tools like NetworkX (Python) to explore graph isomorphism.<br>• Collaborative exercises on Euler/Hamiltonian paths and planar graph identification. | 09 |
| 5 | **Trees:** Definitions, Properties and Examples, Rooted trees, trees and sorting weighted trees and prefix Codes.<br>**Self-Learning and Team Work*:**<br>• Practice tree traversals (preorder, postorder) with programming tasks.<br>• Learn about applications in data compression (prefix codes) individually. | 09 |

| | | |
|---|---|---|
| | • Team-based exploration of weighted trees for optimization problems (like Huffman coding). | |

**NOTE: *Self-Learning and Term Work are evaluated only through continuous internal evaluation (CIE) and will not be assessed during the Semester End Examination (SEE).**

**Text Book:**

| Sl. No. | Author/s | Title | Publisher Details |
|---|---|---|---|
| 1 | Ralph. P. Grimaldi, B.V. Ramana | Discrete and Combinatorial Mathematics | 5th Edition, Pearson Education, 2013. |

**Reference Books:**

| Sl. No. | Author/s | Title | Publisher Details |
|---|---|---|---|
| 1 | Kenneth. H. Rosen | Discrete Mathematical Structures Theory and Application | 5th Edition, PHI/Pearson, Education, 2004. |
| 2 | Kolman, Busby and Ross | Discrete Mathematical Structures | 4th Edition, Prentice Hall of India Pvt. Ltd, 2009. |
| 3 | Dr. D.S. Chandrashekar | Discrete Mathematical Structures | 4thEdition Prism Books Private Limited, 2013. |
| 4 | Dr. S.K. Yadav | Discrete Mathematical Structures | Ane Books Pvt. Ltd. |

**Web Resources:**

| Sl. No. | Web Links |
|---|---|
| 1 | https://nptel.ac.in/courses/106/106/106106094/ |
| 2 | https://nptel.ac.in/courses/106106183/ |

**Course Articulation Matrix:**

| Course Outcomes | Program Outcomes | | | | | | | | | | | Program Specific Outcomes | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 | PSO3 |
| CO1 | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 3 | - | 3 |
| CO2 | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 3 | - | 3 |
| CO3 | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 3 | - | 3 |
| CO4 | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 3 | - | 3 |
| CO5 | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 3 | - | 3 |

High – 3, Medium – 2, Low – 1

| DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING | | | | | | |
|---|---|---|---|---|---|---|
| *Course Code: 24CS36L* | *Course Title: Data Structures Lab* | | | | | |
| *Total Credits: 1.5 L-0: T-0: P-45* | *SDGs addressed: 4,9,11* | | *Course Category: PCCL* | *Course Type: Laboratory* | | |
| *Teaching Learning Process* | | *Total Contact Hours/semester* | *Assessment Weightage and Marks* | | | |
| | *Lecture* | - | | **CIE** | **SEE** | **Total** |
| | *Tutorial* | - | *Weightage, %* | *40* | *60* | *100* |
| | *Practical* | 45 | *Maximum Marks* | *40* | *60* | *100* |
| | *Self-learning and Term Work* | - | *Minimum Marks* | *20* | *25* | *45** |
| | *Total* | 45 | | | | |

**Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)**

**Course Prerequisite: C programming.**

**Course Objectives:**

| Sl. No. | Course Objectives |
|---|---|
| 1 | Gain knowledge in practical applications of data structures and to introduce various techniques for representation of the data in the real world. |
| 2 | Strengthening the ability of the students to identify and apply the suitable data structure for the given real-world problem. |
| 3 | Develop skills to design and analyze simple linear and nonlinear data structures. |

**Course Outcomes (COs)**

| CO | Course Outcomes | Highest Level of Cognitive Domain | Working Knowledge |
|---|---|---|---|
| CO1 | Implement programs to illustrate the use of pointers to arrays, structures and functions. | L3 | WK2, WK6 |
| CO2 | Implement various operations on Linear lists with Array and Linked list representations and analyze the performance of operations. | L3 | WK2, WK4 |
| CO3 | Implement different data structures such as Stack, Queue, Tree and Graph using Array and Linked list representations with static or dynamic memory allocation and analyze the performance of operations, use stack for expression conversion and evaluation. | L3 | WK2, WK4 |

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

**Course Content / Syllabus:**

| Weeks | List of Programs | No. of Hours |
|---|---|---|
| 1 | Write programs for the following to illustrate the use of pointers. | 03 |

| | | |
|---|---|---|
| | a) Create 3 pointers called P1, P2, P3 and an integer variable X where P1 points to P2, P2 points to P3 and P3 points to X. Demonstrate how to read and retrieve the value of X using P1, P2 and P3.<br>b) Crate a pointer to an integer array, Read and display the contents using pointer and also display the contents in reverse order using pointer.<br>c) Create a pointer to an integer array and pass this as parameter to a function called Reverse which reverses the array contents using pointer. Write different functions called Read (P) to read the array elements using pointer P and Display (P) to display the array contents using P before and after reverse.<br>d) Create a pointer to two-dimensional array of integers. Write functions Read (P) and Display (P) to read and display the contents using pointer.<br>e) Create two pointers called PS1 and PS2 to two strings, Read the contents using pointers and also write a function StrCat (PS1, PS2) which concatenates second string to the end of first string using pointers PS1 and PS2. | |
| 2 | Define a structure called Student with the members: Name, Reg_no, marks in 3 tests and average_ marks.<br>Develop a menu driven program to perform the following by writing separate function for each operation: a) read information of N students b) display student's information c) to calculate the average of best two test marks of each student.<br>Note: Allocate memory dynamically and illustrate the use of pointer to an array of structure. | 03 |
| 3 | Define a structure called Time containing 3 integer members (hour, minute, second). Develop a menu driven program to perform the following by writing separate function for each operation.<br>a) Read (T): To read time b) Display (T):To display time c) update(T):To Update time d) Add (T1, T2) : Add two times.<br>Update function increments the time by one second and returns the new time (if the increment results in 60 seconds, then the second member is set to zero and minute member is incremented by one. If the result is 60 minutes, the minute member is set to zero and the hour member is incremented by one. Finally, when the hour becomes 24, Time should be reset to zero. While adding two time variable, normalize the resultant time value as in the case of update function.<br>Note: Illustrate the use of pointer to pass time variable to different functions. | 03 |
| 4 | Develop a menu driven program to implement the following operations on array storage representation with static and dynamic memory allocation.<br>i)Insert by position<br>ii) Delete by position<br>iii)Insert by key<br>iv) Delete by key<br>v) Insert by order<br>vi) Search by key | 03 |

| | | |
|---|---|---|
| | vii) Search by position<br>viii) Reverse the contents. | |
| 5 | Develop a menu driven program to implement singly linked list with various operations such as<br>i) Insertion and Deletion at front/rear<br>ii) Insertion and Deletion at the specified position<br>iii) Delete by Key<br>iv) Search by key<br>v) Create an ordered list<br>vi) Reverse a list<br>vii) Creating a copy of the list | 03 |
| 6 | Develop a menu driven program to implement Circular singly linked list with various operations such as (Note: Use Header node to simplify he complex operations)<br>i) Insertion and Deletion at front/rear<br>ii) Insertion and Deletion at the specified position<br>iii) Delete by Key<br>iv) Search by key<br>v) Create an ordered list<br>vi) Reverse a list<br>vii) Creating a copy of the list | 03 |
| 7 | Develop a menu driven program to implement Circular Double linked list with Header node to perform various operations such as<br>i) Insertion and Deletion at front/rear<br>ii) Insertion and Deletion at the specified position<br>iii) Delete by Key<br>iv) Search by key<br>v) Create an ordered list<br>vi) Reverse a list and<br>vii) Creating a copy of the list | 03 |
| 8 | Develop a menu driven program to implement Stack with static and dynamic memory allocation mechanisms. (Represent Stack using structure) | 03 |
| 9 | a) Develop a menu driven program to convert infix expressions to postfix expression and evaluate the postfix expression.<br>b) Develop a menu driven program to convert infix expression to prefix and evaluate the prefix expression. | 03 |
| 10 | a) Develop a menu driven program to implement ordinary Queue with static and dynamic memory allocation mechanisms. (Represent Queue using structure).<br>b) Develop a menu driven program to implement Circular Queue with static and dynamic memory allocation mechanisms.(Represent Queue using structure) | 03 |
| 11 | a) Develop a menu driven program to implement Double Ended Queue with static and dynamic memory allocation mechanisms. (Represent Queue using structure) | 03 |

| | | |
|---|---|---|
| | b) Develop a menu driven program to implement Priority Queue with static and dynamic memory allocation mechanisms (Represent Queue using structure) | |
| 12 | a) Develop a menu driven program to implement binary search tree and traversal techniques.<br>b) Develop a menu driven program to implement Graph traversal techniques. | 03 |
| 13 | **Revision** | 03 |
| 14 | **Revision** | 03 |
| 15 | **Laboratory Test** | 03 |

**Reference Books:**

| Sl. No. | Author/s | Title | Publisher Details |
|---|---|---|---|
| 1 | Ellis Horowitz and Sartaj Sahni | Fundamentals of Data Structures in C | 2nd Edition, Universities Press, 2014. |
| 2 | Richard F. Gilberg and Behronz A. Forouzan | Data structures, A Pseudo code Approach with C | 2nd Edition, Universities Press, 2014. |
| 3 | Seymour Lipschutz | Data Structures | Revised 1st edition, McGrawHill 2014. |
| 4 | Yedidyah, Augenstein, Tannenbaum | Data Structures Using C | 1st Edition, Pearson Education, 2019. |

**Web Resources:**

| Sl. No. | Web Links |
|---|---|
| 1 | https://nptel.ac.in/courses/106/105/106105085/ |
| 2 | https://nptel.ac.in/courses/106/103/106103069/ |

**Course Articulation Matrix:**

| Course Outcomes | Program Outcomes | | | | | | | | | | | Program Specific Outcomes | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 | PSO3 |
| CO1 | 3 | 3 | 3 | - | - | - | - | - | 3 | 3 | - | 3 | - | 3 |
| CO2 | 3 | 3 | 3 | 3 | - | - | - | - | 3 | 3 | - | 3 | - | 3 |
| CO3 | 3 | 3 | 3 | 3 | - | - | - | - | 3 | 3 | - | 3 | - | 3 |

High – 3, Medium – 2, Low – 1

| DEPARTMENT: HUMANITIES | | | | | | |
|---|---|---|---|---|---|---|
| *Course Code:* 24HU311 | *Course Title:* ***Universal Human Values - II*** | | | | | |
| ***Total Credits: 02 L-30: T-0: P-0: SL and TW-30*** | ***SDGs addressed: 3,4,5,6,7,8,9,10,12,13*** | ***Course Category: HSMC*** | ***Course Type: Theory*** | | | |
| ***Teaching Learning Process*** | | ***Total Contact Hours/semester*** | ***Assessment Weightage and Marks*** | | | |
| | ***Lecture*** | **30** | | **CIE** | **SEE** | **Total** |
| | ***Tutorial*** | **-** | ***Weightage, %*** | ***40*** | ***60*** | ***100*** |
| | ***Practical*** | **-** | ***Maximum Marks*** | ***20*** | ***30*** | ***50*** |
| | ***Self-learning and Term work*** | **30** | ***Minimum Marks*** | ***10*** | ***13*** | ***23**** |
| | ***Total*** | **60** | | | | |

**Note: *For passing the student must score a Minimum of 23 Marks (CIE+SEE: 10 + 13 or 11 + 12)**

**Course Pre-Requisite:** Students Induction Program (desirable).

**Course Objectives:**

1. Development of a holistic perspective based on self-exploration about themselves (human being), family, society and nature/existence.
2. Understanding (or developing clarity) of the harmony in the human being, family, society and nature/existence
3. Strengthening of self-reflection.
4. Development of commitment and courage to act.

**Course Outcomes (COs)**

| CO | Course Outcomes | Highest Level of Cognitive Domain | Working Knowledge |
|---|---|---|---|
| CO1 | Understand self (I) clearly through self-exploration and demonstrate natural acceptance in situations by right understanding. | L2 | WK1, WK3 |
| CO2 | Distinguish between self (I) body, and co-existence of both to be in harmony. | L2 | WK2, WK6 |
| CO3 | Relate to family and society by understanding nine universal human values in human- human relationship for harmony and prosperity. | L2 | WK1, WK4 |
| CO4 | Explain interconnectedness and mutual fulfilment at all four orders of nature for co- existence. | L2 | WK7, WK8 |
| CO5 | Develop commitment for holistic understanding of harmony by natural acceptance of human values in his/her profession and society. | L2 | WK5, WK8 |

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

**Course Content / Syllabus:**

| Unit No. | Content | Lecture Hours |
|---|---|---|
| 1 | **Introduction - Need, Basic Guidelines, Content and Process for Value Education**<br>1. Purpose and motivation for the course, recapitulation from Universal Human Values-I.<br>2. Self-Exploration–what is it? - Its content and process; 'Natural Acceptance' and Experiential Validation- as the process for self-exploration.<br>3. Continuous Happiness and Prosperity- A look at basic Human Aspirations.<br>4. Right understanding, Relationship and Physical Facility- the basic requirements for fulfilment of aspirations of every human being with their correct priority.<br>5. Understanding Happiness and Prosperity correctly- A critical appraisal of the current scenario.<br>6. Method to fulfill the above human aspirations: understanding and living in harmony at various levels.<br>**Self-Learning and Term work**<br>• Identify one personal decision influenced by natural acceptance and note its effect on your choice.<br>• Identify a campus or community issue and propose a simple solution based on harmony principles. | 06 |
| 2 | **Understanding Harmony in the Human Being - Harmony in Myself!**<br>7. Understanding human being as a co-existence of the sentient 'I' and the material 'Body'.<br>8. Understanding the needs of Self ('I') and 'Body' - happiness and physical facility.<br>9. Understanding the Body as an instrument of 'I' (I being the doer, seer and enjoyer).<br>10. Understanding the characteristics and activities of 'I' and harmony in 'I'.<br>11. Understanding the harmony of I with the Body: Sanyam and Health; correct appraisal of Physical needs, meaning of Prosperity in detail.<br>12. Programs to ensure Sanyam and Health.<br>**Self-Learning and Term work**<br>• Reflect on your daily routine and identify 2–3 habits that promote harmony between your mind ('I') and body, noting their impact on well-being.<br>• Design a simple weekly wellness plan (eg. exercises-self and body, food- self and body). | 06 |
| 3 | **Understanding Harmony in the Family and Society- Harmony in Human-Human Relationship**<br>13. Understanding values in human-human relationship; meaning of Justice (nine universal values in relationships) and program for its | 06 |

| | | |
|---|---|---|
| | fulfilment to ensure mutual happiness; Trust and Respect as the foundational values of relationship.<br>14. Understanding the meaning of Trust; Difference between intention and competence<br>15. Understanding the meaning of Respect, Difference between respect and differentiation; the other salient values in relationship.<br>16. Understanding the harmony in the society (society being an extension of family): Resolution, Prosperity, fearlessness (trust) and co-existence as comprehensive Human Goals<br>17. Visualizing a universal harmonious order in society- Undivided Society, Universal Order- from family to world family.<br>**Self-Learning and Term work**<br>• Reflect on a recent interaction with peers or family and identify how trust, respect, or other universal values influenced the outcome.<br>• Prepare a flowchart or concept map showing how values in family can scale to society and the wider world. | |
| 4 | **Understanding Harmony in the Nature and Existence - Whole existence as Coexistence**<br>18. Understanding the harmony in the Nature.<br>19. Interconnectedness and mutual fulfilment among the four orders of nature- recyclability and self-regulation in nature.<br>20. Understanding Existence as Co-existence of mutually interacting units in all pervasive space.<br>21. Holistic perception of harmony at all levels of existence.<br>**Self-Learning and Term work**<br>• Observe one aspect of your daily life (water, energy, waste, plants, or animals) and note how your actions impact other orders of nature.<br>• Create a diagram or model showing the interdependence of human, animal, plant, and material orders, and discuss ways to maintain balance. | 06 |
| 5 | **Implications of the above Holistic Understanding of Harmony on Professional Ethics**<br>22. Natural acceptance of human values.<br>23. Definitiveness of Ethical Human Conduct.<br>24. Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order.<br>25. Competence in professional ethics: a. Ability to utilize the professional competence for augmenting universal human order b. Ability to identify the scope and characteristics of people friendly and eco-friendly production systems, c. Ability to identify and develop appropriate technologies and management patterns for above production systems.<br>26. Case studies of typical holistic technologies, management models and production systems.<br>27. Strategy for transition from the present state to Universal Human Order: a. At the level of individual: as socially and ecologically | 06 |

| | | |
|---|---|---|
| | responsible engineers, technologists and managers b. At the level of society: as mutually enriching institutions and organizations.<br>28. Sum up.<br>**Self-Learning and Term work**<br>• Reflect on a recent project or assignment and identify one way your technical or professional skills could contribute to eco-friendly or people-friendly solutions.<br>• Identify one campus or community process (e.g., energy use, waste management) and propose a simple, value-based strategy to improve sustainability and social responsibility. | |

**Text Books:**

- Human Values and Professional Ethics by R R Gaur, R Sangal, G P Bagaria, Excel Books, New Delhi, 2010.

**References**

- Jeevan Vidya: Ek Parichaya, A Nagaraj, Jeevan Vidya Prakashan, Amarkantak, 1999.
- Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
- The Story of Stuff (Book).
- The Story of My Experiments with Truth - by Mohandas Karamchand Gandhi
- Small is Beautiful - E. F Schumacher.
- Slow is Beautiful - Cecile Andrews
- Economy of Permanence - J C Kumarappa
- Bharat Mein Angreji Raj - PanditSunderlal
- Rediscovering India - by Dharampal
- Hind Swaraj or Indian Home Rule - by Mohandas K. Gandhi
- India Wins Freedom - Maulana Abdul Kalam Azad
- Vivekananda - Romain Rolland (English)
- Gandhi - Romain Rolland (English)

**Weblinks:**

- https://onlineethics.org/

**Course Articulation Matrix:**

| Course Outcomes | Program Outcomes | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
| CO1 | - | - | - | - | - | 1 | 2 | 2 | - | - | - |
| CO2 | - | - | - | - | - | 1 | 1 | 1 | - | - | - |
| CO3 | - | - | - | - | - | 2 | 2 | 2 | - | - | - |
| CO4 | - | - | - | - | - | 2 | 2 | 2 | - | - | - |
| CO5 | - | - | - | - | - | 2 | 2 | 2 | - | - | - |

High – 3, Medium – 2, Low – 1

## EVALUATION FRAMEWORK FOR COURSES IN THE 160 CREDIT SCHEME

**Evaluation Scheme for Theory course with 3 or 4 credits**

**Continuous Internal Evaluation – CIE**

| Event | Event Type | Marks Allotted | Duration |
|---|---|---|---|
| Event-I of CIE (Test-1) | Written Test | 30 | 1 Hour 15 minutes |
| Event-II of CIE | Event | 20 | As per calendar of events |
| Event-III of CIE Test-2 | Written Test | 30 | 1 Hour 15 minutes |

**Note:**

1. Both the written tests, Test-1 and Test-2 shall be conducted for 30 marks each in 1 hour 15 minutes duration and the Event-II shall be conducted for 20 marks.
2. The Event-II shall be skill-based assessment such as Assignment/ Technical presentation/ Case study / hands-on activity / Mini projects / Sci-tech activity / Data analysis or any other events as decided by the course teacher.
3. The total of 80 marks shall be reduced to 40 marks for CIE.
4. A student must score a minimum of 50% i.e., 20 marks out of 40 from all the events (I, II and III) to gain the eligibility to appear for SEE in addition to attendance requirements.
5. A make-up test (for Event-I or Event-III) shall be given to improve the CIE marks.

**Semester End Examination (SEE)**

| Exam | Event Type | Maximum Marks | Duration |
|---|---|---|---|
| SEE | Written Examination | 100 | 3 Hours |

**Note:**

1. SEE shall be conducted for 100 marks and the marks scored shall be reduced to 60 marks. A student must score a minimum of 42 marks out of 100 (24 marks out of 60) in the SEE. However, to obtain a passing grade in a course, the student has to score a minimum of 45 marks (CIE+SEE: 20 + 25 or 21 + 24).
2. SEE Question paper shall be set for 100 marks and shall have two parts (Part-A and Part-B). Questions under Part-A are compulsory and questions under Part-B shall have internal choice.
3. All COs shall be covered in both Part-A and Part-B and preferably have equal weightage.

## Evaluation Scheme for Theory course with 1 or 2 credits

**Continuous Internal Evaluation – CIE**

| Event | Event Type | Marks Allotted | Duration |
|---|---|---|---|
| Event-I of CIE (Test-1) | Written Test | 15 | Minimum 45 minutes |
| Event-II of CIE | Event | 10 | - |
| Event-III of CIE Test-2 | Written Test | 15 | Minimum 45 minutes |

**Note:**

1. Both the written tests, Test-1 and Test-2 shall be conducted for 15 marks each in minimum 45 minutes duration and the Event-II shall be conducted for 10 marks.
2. The Event-II shall be skill-based assessment such as Assignment/ Technical presentation/ Case study / hands-on activity / Mini projects / Sci-tech activity / Data analysis or any other events as decided by the course teacher.
3. The total of 40 marks shall be reduced to 20 marks for CIE.
4. A student must score a minimum of 50% i.e., 10 marks out of 20 from all the events (I, II and III) to gain the eligibility to appear for SEE in addition to attendance requirements.
5. A make-up test (for Event-I or Event-III) shall be given to improve the CIE marks.

**Semester End Examination (SEE)**

| Exam | Event Type | Maximum Marks | Duration |
|---|---|---|---|
| SEE | Written Examination | 60 | 2 Hours |

**Note:**

1. SEE shall be conducted for 60 marks and the marks scored shall be reduced to 30 marks. A student must score a minimum of 24 marks out of 60 in the SEE. However, to obtain a passing grade in a course, the student has to score a minimum of 23 marks (CIE+SEE: 10 + 13 or 11 + 12).
2. SEE Question paper shall be set for 60 marks and shall have two parts (Part-A and Part-B). Questions under Part-A are compulsory for 30 and questions under Part-B shall have internal choice for 30 marks.
3. All COs shall be covered and preferably have equal weightage.

**Evaluation Scheme for Theory with Integrated Laboratory Course:**

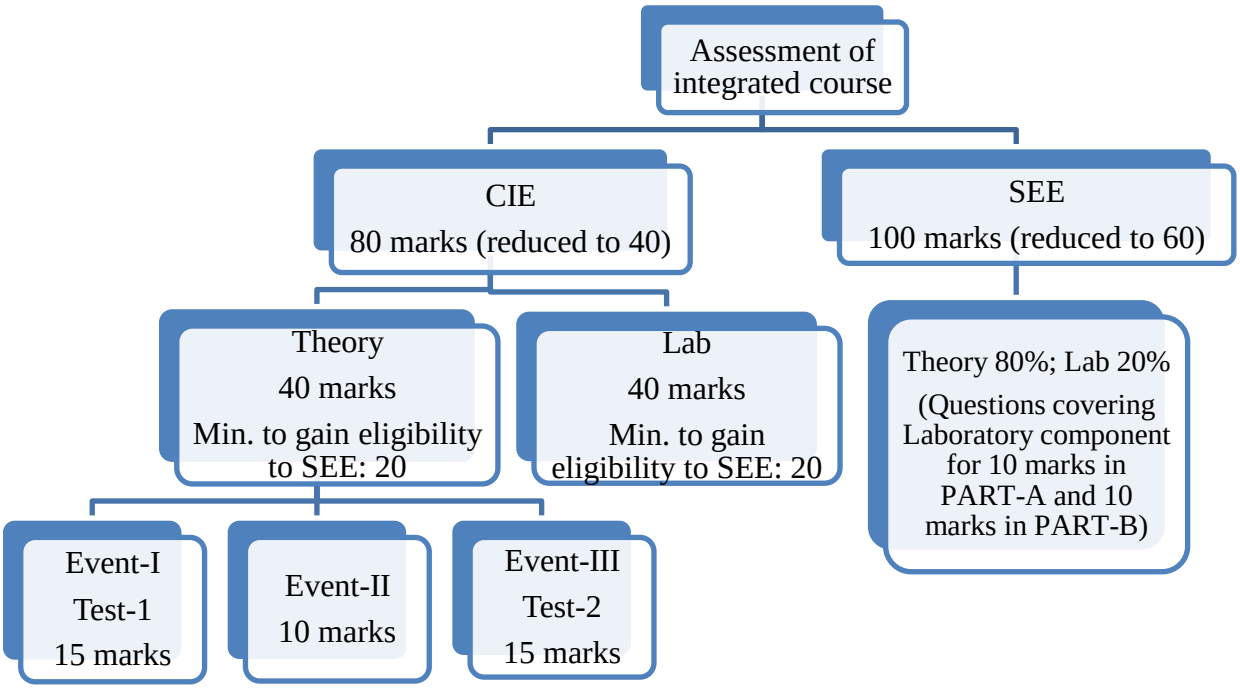
Assessment for integrated course is to be conducted in two parts: CIE and SEE. A total of 40 marks is to be assessed through CIE which includes theory and laboratory components.

Theory component involves three events viz., Event-I (of CIE) to be conducted as Test-1 for a total of 15 marks, Event-II of CIE is at the discretion of the course teacher which may be assignment, quiz, seminar or poster presentation, model, role play, etc assessed for a total of 10 marks and Event-III (of CIE) is to be conducted as Test-2 for a total of 15 marks. Together, the theory component contributes to a total of 40 marks which shall eventually be reduced to 20 marks. Laboratory component is assessed for a total of 40 marks which shall eventually be reduced to 20 marks. A student must secure a minimum of 20 marks out of 40 marks in theory component and a minimum of 20 marks out of 40 marks in the lab component to be eligible for Semester End Examination (SEE).

**Continuous Internal Evaluation for theory component– CIE**

| Event | Event Type | Marks Allotted | Duration |
|---|---|---|---|
| Event-I of CIE (Test-1) | Written Test | 15 | Minimum 45 minutes |
| Event-II of CIE | Event | 10 | - |
| Event-III of CIE Test-2 | Written Test | 15 | Minimum 45 minutes |

The SEE shall consist of questions from both theory and laboratory content for a total of 150 marks. The SEE question paper shall consist of questions on theory for a total 120 marks and questions on laboratory for a total of 30 marks. The SEE question paper shall have questions for 10 marks both in PART-A and PART-B (with internal choice) covering laboratory components.




**Evaluation Scheme for Theory Course with zero or 1 credit without SEE:**

**Continuous Internal Evaluation – CIE**

| Event | Event Type | Marks Allotted | Duration |
|---|---|---|---|
| Event-I of CIE (Test-1) | Written Test | 30 | 1 Hour 15 minutes |
| Event-II of CIE | Event | 20 | As per calendar of events |
| Event-III of CIE Test-2 | Written Test | 30 | 1 Hour 15 minutes |

**Note:**

1. Both the written tests, Test-1 and Test-2 shall be conducted for 30 marks each in 1 hour 15 minutes duration and the Event-II shall be conducted for 20 marks.
2. The Event-II shall be skill-based assessment such as Assignment/ Technical presentation/ Case study / hands-on activity / Mini projects / Sci-tech activity / Data analysis or any other events as decided by the course teacher.
3. The total of 80 marks shall be reduced to 40 marks for CIE.
4. A student must score a minimum of 50% i.e., 20 marks out of 40 from all the events (I, II and III) to gain the eligibility to appear for SEE in addition to attendance requirements.
5. A make-up test (for Event-I or Event-III) shall be given to improve the CIE marks.

## Evaluation Scheme for Practical/Laboratory Course of 1.5 credits:

**Continuous Internal Evaluation – CIE**

| Event | Event Type | Marks Allotted | Duration |
|---|---|---|---|
| CIE – 1 | Experiments / Programs / Models | 30 | 3 Hours/session |
| CIE – 2 | Internal Test | 10 | 3 Hours |

**Note:**

1. The experiments / program / models shall preferably be designed to be conducted for three hours in each laboratory session.
2. A student shall perform all the experiments / programs / models in a semester by performing experiments / executing programs / preparing models as scheduled and the student shall be evaluated every week for each practical session.-
3. The marks scored from experiments / programs / models shall be reduced proportionately to 30 marks at the end of the semester.
4. An internal test in the 13$^{th}$ week of the semester shall be conducted after the completion of all the practical sessions as per the procedures mentioned in the evaluation policy and the marks secured shall be reduced proportionately to 10 marks.
5. The sum of marks scored from experiments / programs / models for 30 marks and from the internal test for 10 marks shall be the final CIE marks.
6. A student must score a minimum of 20 marks out of 40 to gain the eligibility for SEE in addition to attendance requirements.

**Rubrics for Continuous Assessment Evaluation of Laboratory Course**

Laboratory courses will be evaluated under criteria namely Preparedness, Conduction & Result Interpretation and Result Recording & Report Writing. At the end of the course a laboratory test will be conducted where student will have to demonstrate an experiment that is assigned randomly. The evaluation rubrics is shown in the Table below showing recording of semester end test and final CIE computation.

| Criteria | Excellent (10 Marks) | Good (7 marks) | Satisfactory (5 marks) | Poor(0) |
|---|---|---|---|---|
| **C1** Preparedness (2 Marks) | With adequate information, necessary preparation and analytical skills: Programs for the problem are written neatly in the observation book with clearly defined inputs and the expected outputs. Able to explain the logic | With limited information, necessary preparation and analytical skills. Programs for the problems are written neatly in the observation book with clearly defined inputs and the expected outputs but not able to explain the logic. | Only Programs for the problems are written without adequate information, and analytical skills. | Student is present with no preparation |
| **C2** Conduction & Result Interpretation (5 Marks) | Executes the programs and demonstrates the results in an excellent manner and able to modify. | Executes the programs and demonstrates the results effectively, but not able to modify. | Executes the programs, but not able to demonstrate the results effectively. | Not able to execute. |

| | | | | |
|---|---|---|---|---|
| **C3** Result Recording & Report writing (3 Marks) | Submits the report with excellent inference. | Submits the report with good inference. | Submits the report with moderate inference. | Not submitted |

**Continuous Assessment with Preparedness, Conduction & Result Interpretation, and Result Recording and Report writing: CA1 (30 Marks)**

| Sl No. | Student Reg No | Student Name | Lab1 | | | Lab2 | | | …….. | | | Lab12 | | | Average CA1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | C1 | C2 | C3 | C1 | C2 | C3 | C1 | C2 | C3 | C1 | C2 | C3 | 30 Marks |
| 1 | | | | | | | | | | | | | | | |
| 2 | | | | | | | | | | | | | | | |
| 3 | | | | | | | | | | | | | | | |

**Continuous Internal Evaluation of the performance in practical classes (CIE-1: 12 Programs)**

| Sl No. | Student Reg No | Student Name | CA2: Conduction & Result Interpretation |
|---|---|---|---|
| | | | 10 marks |
| 1 | | | |
| 2 | | | |
| 3 | | | |
| . | | | |
| . | | | |

**Semester End Practical Test (CIE-2: Internal Test)**

**Semester End Examination – SEE**

| Event | Event Type | Marks Allotted | Duration |
|---|---|---|---|
| SEE | Experiments / Programs / Models | 100 | 3 Hours |

1. SEE shall be conducted for 100 marks in a duration of 3 hours and the marks scored shall be reduced to 60 marks.
2. A student must score a minimum of 24 marks out of 60 marks in the SEE. However, the total passing marks in the laboratory courses is 45 out of 100 (CIE + SEE: 20 + 25 or 21 + 24).
3. All COs shall be covered and preferably have equal weightage.